

Lifetime cancer prevalence and life history traits in mammals.

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Cancer is a common diagnosis in many mammalian species, yet they vary in their vulnerability to cancer. The factors driving this variation are unknown, but life history theory offers potential explanations to why cancer defense mechanisms are not equal across species. Here we report the prevalence of neoplasia and malignancy in 37 mammalian species, representing 11 mammalian orders, using 42 years of well curated necropsy data from the San Diego Zoo and San Diego Zoo Safari Park. We collected data on life history components of these species and tested for associations between life history traits and both neoplasia and malignancy, while controlling for phylogenetic history. These results support Peto's paradox, in that we find no association between lifespan and/or body mass and the prevalence of neoplasia or malignancy. However, a positive relationship exists between litter size and prevalence of malignancy ($P = 0.005$, Adj. $R^2 = 0.212$), suggesting that a species' life history strategy may influence cancer vulnerabilities. Lastly, we tested for the relationship between placental invasiveness and malignancy. We find no evidence for an association between placental depth and malignancy prevalence ($P = 0.618$, Adj. $R^2 = 0.068$). Life history theory offers a powerful framework to understand variation in cancer defenses across the tree of life. These findings provide insight into the relationship between life history traits and cancer vulnerabilities, which suggest a trade-off between reproduction and cancer defenses. Why are some mammals more vulnerable to cancer than others? We test whether life history trade-offs may explain this variation in cancer risk. Bigger, longer-lived animals do not develop more cancer compared to smaller, shorter-lived animals. However, we find a positive association between litter size and cancer prevalence in mammals.

Elephant APOBEC3A cytidine deaminase induces massive double-stranded DNA breaks and apoptosis.

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The incidence of developing cancer should increase with the body mass, yet is not the case, a conundrum referred to as Peto's paradox. Elephants have a lower incidence of cancer suggesting that these animals have probably evolved different ways to protect themselves against the disease. The paradox is worth revisiting with the realization that most mammals encode an endogenous APOBEC3 cytidine deaminase capable of mutating single stranded DNA. Indeed, the mutagenic activity of some APOBEC3 enzymes has been shown to introduce somatic mutations into genomic DNA. These enzymes are now recognized as causal agent responsible for the accumulation of CG->TA transitions and DNA breaks leading to chromosomal rearrangements in human cancer genomes. Here, we identified an elephant A3Z1 gene, related to human APOBEC3A and showed that it could efficiently deaminate cytidine, 5-methylcytidine and produce DNA breaks leading to massive apoptosis, similar to other mammalian APOBEC3A enzymes where body mass varies by up to four orders of magnitude. Consequently, it could be considered that eAZ1 might contribute to cancer in elephants in a manner similar to their proposed role in humans. If so, eAZ1 might be particularly well regulated to counter Peto's paradox.

Cancer Prevalence Across Vertebrates.

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Cancer is pervasive across multicellular species, but what explains differences in cancer prevalence across species? Using 16,049 necropsy records for 292 species spanning three clades (amphibians, sauropsids and mammals) we found that neoplasia and malignancy prevalence increases with adult weight (contrary to Peto's Paradox) and somatic mutation rate, but decreases with gestation time. Evolution of cancer susceptibility appears to have undergone sudden shifts followed by stabilizing selection. Outliers for neoplasia prevalence include the common porpoise (<1.3%), the Rodrigues fruit bat (<1.6%) the black-footed penguin (<0.4%), ferrets (63%) and opossums (35%). Discovering why some species have particularly high or low levels of cancer may lead to a better understanding of cancer syndromes and novel strategies for the management and prevention of cancer.

Factors related with the time to surgery in waiting-list patients for knee prostheses.

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In the year 2006 the Hospital Complex of Jaén began a new model of attention for consultation of knee osteoarthritis (CMAR), to offer quality treatment to the patient based on clinical practice guidelines. The objective of this study was to analyse the efficiency of the unit, in terms of delaying the time until a knee prosthesis is needed. An analysis of a retrospective cohort of patients seen at the CMAR was made, with a minimum available data set, and a Kaplan Meier analysis carried out in order to evaluate the time until the referral of the patient to surgery, as well as a Cox regression analysis in order to study the factors related with the time until the referral. 224 patients were included, with mean age of 65.7 years (95% CI 64 to 67), 67.9% women. 48.2% had a stage III affection (Kellgren-Lawrence), 27.5% stage 2, 15.3% stage 4, and 9% stage 1. 90.2% (202) received treatment with viscosupplementation (Hyaluronic acid NASHA). Single joint affection ($P<.0001$; OR 0,267, 95% CI 0.130 to 0,549), a higher degree of osteoarthritis ($P=.048$; OR 0,410, 95% CI 0.169 to 0,992), and lower intensity of pain ($P<.0001$; OR 1,091, 95% CI 1.044 to 1,141), were the factors related with shorter time until surgery. Treatment with viscosupplementation increased the time until surgery in all patients. Factors like a lower age of the patient and the involvement of a single joint influenced the referral of the patient to surgery. The administration of viscosupplementation, delayed the time until the patient was finally referred to surgery.

Sustained therapeutic effects of self-assembled hyaluronic acid nanoparticles loaded with α -Ketoglutarate in various osteoarthritis stages.

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Osteoarthritis (OA) is a prevalent degenerative disease characterized by irreversible destruction of articular cartilage, for which no current drugs are known to modify its progression. While intra-articular (IA) injections of hyaluronic acid (HA) offer temporary relief, their effectiveness and long-term benefits are debated. Alpha-ketoglutarate (α KG) has potential chondroprotective properties, but its use is limited by a short half-life and poor cartilage-targeting efficiency. Here, we developed self-assembled HA- α KG nanoparticles (NPs) to combine the benefits of both HA and α KG, showing stability, bioavailability, and sustained pH-responsive release in the knee joint. In both early and advanced OA stages in mice, HA, α KG, and HA- α KG NPs could relieve pain, enhance mobility, and reduce cartilage damage, with HA- α KG NPs demonstrating the best efficacy. Mechanistically, α KG not only promotes cartilage matrix synthesis but also inhibits degradation by activating the PERK-ATF4 signaling pathway to reduce endoplasmic

reticulum stress (ERS) in chondrocytes. This study highlights the therapeutic potential of HA- α KG NPs for treating various OA stages, with efficient and sustained effects, suggesting rapid clinical adoption and high acceptability among clinicians and patients.

Different doses of hyaluronic acid injections in patients with knee osteoarthritis: A multicenter, randomized, prospective, single-blind, clinical study.

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Many formulations and dosing regimens are available for hyaluronic acid (HA). To compare different doses of linear, high-molecular weight (HMW) HA injections among patients with knee osteoarthritis (OA). Hundred patients were included in this randomized, single-blinded trial and randomly divided into three HA injection groups. The first group received five weekly 20 mg HA injections, the second group received three weekly 32 mg HA injections, and the third group received a single 48 mg HA injection. Patients were evaluated at baseline, 1, 3, and 6 months after the last injection for pain, stiffness, and function using the visual analog scale (VAS) and the Western Ontario and McMaster Universities Arthritis Index (WOMAC) questionnaire. Quality of life, patient's global assessment, and Timed Up and Go (TUG) test scores were also evaluated. There was significant improvement in the WOMAC, VAS-pain, quality of life, patient's global assessment, and TUG test mean scores at all follow-up time points ($p < 0.001$). However, the groups showed no significant differences in WOMAC, VAS-activity pain, and patient global scores at any follow-up point. Intra-articular injections of different doses of linear HMW HA can improve pain, stiffness, function, and quality of life in patients suffering from knee OA over a six-month period.

a comparison of intra-articular hyaluronic acid and platelet-rich plasma for knee osteoarthritis: a systematic review.

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Knee osteoarthritis (KOA), the most common chronic degenerative condition in an older population, accounts for many disabilities around the world. One of the most popular treatments is intra-articular injection of hyaluronic acid (HA) and platelet-rich plasma (PRP). Prior studies have found that both HA and PRP had a therapeutic effect on KOA. This study aims to perform a systematic review regarding whether PRP is superior to HA for KOA. We conducted a comprehensive literature search using Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines for prospective randomized control trials (pRCTs) in three international databases PubMed, Google Scholar, and ScienceDirect from 2019-2022. Two researchers independently searched the reviews, extracted, and cross-checked the data. The disparity when choosing the literature was resolved by discussion. The modified Jadad was scale used to assess the quality of the included studies. Cochrane risk of bias 2 tool (RoB-2) was used for determininzg risk of bias. Twenty three studies were eligible for inclusion. Four pRCT with the highest Jadad score were selected as best evidence. Risk of bias assesment concluded two studies having a low risk of bias, one is high risk of bias, and the other possesses some concerns.. Three studies found no difference in patient-reported outcomes between PRP and HA group and one study concluded that PRP is more effective than HA in treating KOA. Intra-articular injections of PRP and HA are effective interventions for KOA. However, there is not enough evidence of PRP superiority over HA.

A prospective, randomized, double-blind, placebo controlled study to evaluate the efficacy of intraarticular hyaluronic acid for osteoarthritis of the knee.

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To assess the efficacy of intraarticular (IA) injections of hyaluronic acid (HA) compared to placebo in patients with osteoarthritis (OA) of the knee; and to assess patient satisfaction with treatment relative to placebo, and whether there is a difference between a series of 3 versus 6 consecutive IA injections. We conducted a randomized, double-blind, placebo controlled, 2-arm parallel design trial of 106 patients with radiologically confirmed knee OA. Two-milliliter IA injections using 20 mg/ml HA sodium salt or saline placebo were administered once weekly over 3 weeks (HA and placebo groups), followed by once weekly IA injection with 2.0 ml (20 mg/ml) HA for a further 3 consecutive weeks. The primary efficacy assessment included Western Ontario and McMaster Universities osteoarthritis index (WOMAC) score for knee pain (Week 3 score). Secondary efficacy assessments included WOMAC scores for knee pain at Weeks 6 and 12 (followup), as well as WOMAC stiffness, physical function and quality of life scores, visual analog scale (VAS) scores for pain at rest and following walking and stepping activity, range of knee joint motion, and global patient satisfaction with treatment and quality of life using the SF-36. After 3 weeks of study treatment, both treatment groups showed improvements in knee function, the HA group showing a greater improvement compared to placebo in WOMAC knee pain score ($p < 0.01$). The HA group showed greater ($p < 0.05$) improvement in the overall WOMAC score and VAS pain following walking and stepping activity at Week 3. Results from all other secondary efficacy assessments at Weeks 6 and 12 including patient satisfaction were similar and were not statistically significant between treatment groups, and there were no significant differences between groups for adverse events. Intraarticular HA was superior to placebo in improving knee pain and function, with no difference between 3 or 6 consecutive injections for the primary efficacy assessment.

Hyaluronic acid injections relieve knee pain.

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To evaluate the efficacy of intra-articular viscosupplementation therapy with hyaluronic acid for pain relief of knee osteoarthritis, we conducted a meta-analysis of randomized, double-blinded, placebo-controlled trials. We searched systematically for randomized, double-blinded, placebo-controlled trials of hyaluronic acid (hyaluronan and hylan G-F20) for pain relief of knee osteoarthritis. Studies reporting pain visual analogue scale (VAS) differences were included in the meta-analysis. Changes in pain were measured by VAS for placebo and treatment, and summary estimates of the differences between the 2 arms were calculated at 1 week, 5 to 7 weeks, 8 to 12, and 15 to 22 weeks after the last intra-articular injection. Sources of heterogeneity were assessed using information on quality score, type of viscosupplementation, and VAS change in pain with activity or rest. Heterogeneity across the studies was significant in all analyses ($P < .01$); therefore a random effect model was used. Pain was measured either on activity or at rest. Eleven trials (9 hyaluronan and 2 hylan G-F 20) allowed calculation of the summary estimate of difference in change of VAS pain at 1 week, 6 of the 11 allowed the estimation between 5 to 7 weeks and 8 to 12 weeks, and only 3 at 15 to 22 weeks. The summary estimates of VAS differences between therapy and placebo injection: at 1 week, 4.4 (95% confidence interval [CI], 1.1-7.2); at 5 to 7 weeks, 17.7 (7.5-28.0); at 8 to 12 weeks, 18.1 (6.3-29.9) and at 15 to 22 weeks, 4.4 (-15.3 to 24.1). Intra-articular viscosupplementation was moderately effective in relieving knee pain in patients with osteoarthritis at 5 to 7 and 8 to 10 weeks after the last injection but not at 15 to 22 weeks.

Efficacy of intraarticular hyaluronic acid injections in knee osteoarthritis.

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Intraarticular injections of hyaluronic acid have been advocated for treatment of symptomatic knee osteoarthritis. Appropriate indications and favorable patient response factors, such as radiographic disease severity and age, are not clearly defined for this therapy. The current review of 80 knees with symptomatic osteoarthritis treated with hyaluronic acid revealed that approximately 2/3 of treated knees received 2/3 relief of pain. Hyaluronic acid treatment is not appropriate for all patients with knee osteoarthritis. Overall, less than 50% of treated knees achieved satisfactory results, and only 35% reported increased activity. Twenty-two patients (28% of knees; 22 knees) underwent surgery within 7 months of their index injection, suggesting an inadequate response to treatment. The treatment is not without complication because 11 patients (15% of knees; 12 knees) experienced adverse reactions, including one case of septic arthritis. The authors recommend intraarticular hyaluronic acid only for patients with symptoms and significant surgical risk factors and for patients with mild radiographic disease in whom conservative treatment has failed (physical therapy, weight loss, nonsteroidal antiinflammatory medication, and intraarticular steroid injection). It is inadvisable to treat patients with a complete collapse of joint space or bone loss with intraarticular hyaluronic acid, given their poor clinical response.
